

• CODING WORKBOOK

Java & DSA Interview Coding Workbook

For SDET · 3 Years Experience

Curated coding questions only — no theory, no solutions. Two sections, 22 topics, and 230+ hand-picked problems, ending with must-solve lists, company priorities and a 30-day plan.

230+

Questions

22

Topics

Top 100

Must-Solve

10

Companies

30-Day

Revision Plan

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Solve · Revise · Repeat

Contents

Questions only — no theory, no solutions. Each topic lists its objective, the skills it tests, interview frequency, difficulty, a curated question set, and a recommended solving order. Difficulty is tagged per question: **E** Easy ·

M Medium · **H** Hard.

SECTION 1 • JAVA LOGIC

Numbers	High	Easy-Medium	15 Qs
Patterns	Medium	Easy-Medium	10 Qs
Strings	Very High	Easy-Medium	15 Qs
Arrays	Very High	Easy-Medium	15 Qs
Collections	High	Easy-Medium	12 Qs
Java 8 Streams	Very High	Medium	14 Qs
OOP Coding	High	Medium	10 Qs
Exception Handling	Medium	Easy-Medium	8 Qs
Multithreading	Medium-High	Medium-Hard	10 Qs

SECTION 2 • DSA

Arrays	Very High	Easy-Hard	15 Qs
Strings	High	Easy-Hard	12 Qs
HashMap	Very High	Easy-Medium	10 Qs
Stack	High	Easy-Medium	10 Qs
Queue	Medium	Easy-Medium	8 Qs
Linked List	Very High	Easy-Hard	12 Qs
Trees	Very High	Medium-Hard	12 Qs
Searching	Medium	Easy-Medium	6 Qs
Sorting	Medium	Easy-Medium	8 Qs
Binary Search	High	Medium	8 Qs
Sliding Window	High	Medium	8 Qs
Two Pointer	High	Easy-Medium	8 Qs
Recursion	Medium-High	Medium-Hard	10 Qs

236+

CURATED QUESTIONS

22

TOPICS

100

MUST-SOLVE

30

DAY PLAN

SECTION 1

Java Logic

9 topics · 109 curated questions

Numbers

Patterns

Strings

Arrays

Collections

Java 8 Streams

OOP Coding

Exception Handling

Multithreading

Numbers

HIGH

Easy-Medium

15 Qs

OBJECTIVE

Sharpen numeric reasoning and core math operations.

SKILLS TESTED

Loops, modulo, integer math, conversions

01	Check if a number is prime	E
02	Print all prime numbers up to N	E
03	Reverse an integer without converting to a string	E
04	Check if a number is a palindrome	E
05	Find the factorial of a number (iterative & recursive)	E
06	Print the Fibonacci series up to N terms	E
07	Check if a number is an Armstrong number	E
08	Find the GCD and LCM of two numbers	M
09	Count the number of digits in an integer	E
10	Reduce the sum of digits to a single digit (digital root)	M
11	Check whether a number is a perfect number	M
12	Swap two numbers without a temporary variable	E
13	Find the largest and smallest digit in a number	E
14	Check if a number is a power of two	M
15	Convert a decimal number to binary and back	M

RECOMMENDED ORDER Solve Q1-Q7 (Easy) first to warm up, then Q8-Q15 (Medium).

Patterns

MEDIUM

Easy-Medium

10 Qs

OBJECTIVE

Build fluency with nested loops and control flow.

SKILLS TESTED

Nested loops, indexing, conditionals

01	Print a right-angled triangle of stars	E
02	Print an inverted right-angled triangle	E
03	Print a full pyramid pattern	E
04	Print a diamond pattern	M
05	Print Pascal's triangle	M
06	Print Floyd's triangle	E
07	Print a hollow square pattern	M
08	Print a number pyramid (1 / 12 / 123)	E
09	Print an alphabet pyramid (A / AB / ABC)	E
10	Print a butterfly pattern	M

RECOMMENDED ORDER Follow the list top to bottom; shapes get harder after Q6.

Strings

VERY HIGH

Easy-Medium

15 Qs

OBJECTIVE

Master string traversal and manipulation without heavy libraries.

SKILLS TESTED

char[], StringBuilder, frequency maps

01	Reverse a string without using built-in reverse	E
02	Check if a string is a palindrome	E
03	Count vowels, consonants, digits and spaces	E
04	Find the first non-repeating character	M
05	Count the occurrences of each character	E
06	Check if two strings are anagrams	M
07	Remove duplicate characters from a string	M
08	Find the frequency of each word in a sentence	M
09	Reverse each word in a sentence	M
10	Check if a string contains only digits	E
11	Find the longest word in a sentence	E
12	Convert a string to title case	E
13	Find all permutations of a string	H
14	Compress a string using counts (aaabb → a3b2)	M
15	Check if one string is a rotation of another	M

RECOMMENDED ORDER Q1-Q5 build basics; attempt Q13 (permutations) last.

Arrays

VERY HIGH

Easy-Medium

15 Qs

OBJECTIVE

Traverse and transform arrays, including in-place operations.

SKILLS TESTED

Indexing, two-pass logic, in-place swaps

01	Find the largest and smallest element in an array	E
02	Find the second largest element	M
03	Reverse an array in place	E
04	Find the sum and average of array elements	E
05	Count even and odd numbers	E
06	Remove duplicates from an array	M
07	Find the missing number in an array of 1..N	M
08	Find duplicate elements in an array	M
09	Rotate an array by K positions	M
10	Find the frequency of each element	E
11	Move all zeros to the end	M
12	Find the intersection of two arrays	M
13	Find pairs with a given sum	M
14	Find the maximum subarray sum (Kadane's)	M
15	Merge two sorted arrays	M

RECOMMENDED ORDER Start Q1-Q5; Kadane (Q14) and merge (Q15) are the key medium ones.

Collections

HIGH

Easy-Medium

12 Qs

OBJECTIVE

Apply List, Set, Map and Queue to real coding tasks.

SKILLS TESTED

HashMap, LinkedHashMap, Comparator, iterators

01	Remove duplicates from a List	E
02	Find the frequency of elements using a HashMap	E
03	Sort a HashMap by keys and by values	M
04	Find the first repeated element in a List	M
05	Reverse a List without Collections.reverse	E
06	Merge two Maps and sum values for common keys	M
07	Find common elements between two Lists	E
08	Convert a List into a Map	M
09	Implement an LRU cache using LinkedHashMap	H
10	Find the top K frequent elements	M
11	Sort a list of custom objects using Comparator	M
12	Safely remove entries from a Map while iterating	M

RECOMMENDED ORDER Q9 (LRU) is hard — save it for last.

Java 8 Streams

VERY HIGH

Medium

14 Qs

OBJECTIVE

Solve data problems in functional style with streams.

SKILLS TESTED

map, filter, reduce, collect, groupingBy

01	Find the sum of even numbers in a list using streams	E
02	Find the frequency of each element using groupingBy	M
03	Find the second highest number using streams	M
04	Sort a list of employees by salary using streams	M
05	Group employees by department	M
06	Find the average age of employees per department	M
07	Find duplicate elements in a list using streams	M
08	Convert a list of strings to uppercase using map	E
09	Find the first non-repeating character using streams	M
10	Partition numbers into even and odd	M
11	Count each word in a sentence using streams	M
12	Find the highest-paid employee in each department	M
13	Join a list of strings with a delimiter	E
14	Find the sum of salaries using reduce	M

RECOMMENDED ORDER Master groupingBy (Q2, Q5, Q12) — the most-asked pattern.

OOP Coding

HIGH

Medium

10 Qs

OBJECTIVE

Translate OOP principles and patterns into working code.

SKILLS TESTED

Inheritance, polymorphism, immutability, patterns

01	Design a class hierarchy for shapes with area calculation	M
02	Implement a Bank Account with deposit/withdraw & overdraft rules	M
03	Design a parking lot system (classes & relationships)	H
04	Demonstrate method overloading and overriding	E
05	Design an immutable class	M
06	Implement a thread-safe Singleton	M
07	Implement the Factory pattern for a notification service	M
08	Design an interface-based payment system	M
09	Show composition over inheritance with an example	M
10	Implement equals() and hashCode() correctly for a custom class	M

RECOMMENDED ORDER Design-heavy — Q3 (parking lot) is the stretch problem.

Exception Handling

MEDIUM

Easy-Medium

8 Qs

OBJECTIVE

Write robust code with correct error handling.

SKILLS TESTED

try/catch/finally, custom exceptions, resources

- | | | |
|----|---|---|
| 01 | Create and throw a custom checked exception | E |
| 02 | Create and throw a custom unchecked exception | E |
| 03 | Demonstrate try-with-resources for file handling | M |
| 04 | Show the execution order of try, catch and finally | E |
| 05 | Handle multiple exceptions in a single catch block | E |
| 06 | Re-throw an exception with a wrapped cause | M |
| 07 | Validate input and throw on invalid data | E |
| 08 | Demonstrate exception propagation across method calls | M |

RECOMMENDED ORDER Straightforward — solve in listed order.

Multithreading

MEDIUM-HIGH

Medium-Hard

10 Qs

OBJECTIVE

Solve classic concurrency problems correctly.

SKILLS TESTED

Thread, Runnable, wait/notify, Executors, latches

01	Create threads using Thread and Runnable	E
02	Print odd and even numbers using two threads alternately	H
03	Solve producer-consumer using wait/notify	H
04	Solve producer-consumer using BlockingQueue	M
05	Synchronize a shared counter across threads	M
06	Run parallel tasks using ExecutorService	M
07	Return results from threads using Callable and Future	M
08	Create a deadlock, then resolve it	H
09	Print numbers 1..N sequentially using N threads	H
10	Coordinate threads using CountdownLatch	M

RECOMMENDED ORDER Q2, Q3, Q8, Q9 are the interview favourites — attempt after warm-ups.

SECTION 2

DSA

13 topics · 127 curated questions

Arrays

Strings

HashMap

Stack

Queue

Linked List

Trees

Searching

Sorting

Binary Search

Sliding Window

Two Pointer

Recursion

Arrays

VERY HIGH

Easy-Hard

15 Qs

OBJECTIVE

Core array patterns that appear in nearly every interview.

SKILLS TESTED

Prefix sums, Kadane, intervals, matrix

01	Two Sum	E
02	Best Time to Buy and Sell Stock	E
03	Maximum Subarray (Kadane's)	M
04	Move Zeroes	E
05	Product of Array Except Self	M
06	Find the Duplicate Number	M
07	Merge Intervals	M
08	Rotate Array by K	M
09	Container With Most Water	M
10	Trapping Rain Water	H
11	Maximum Product Subarray	M
12	Missing and Repeating Number	M
13	Majority Element	E
14	Set Matrix Zeroes	M
15	Spiral Matrix	M

RECOMMENDED ORDER Q1-Q4 easy, then medium; Trapping Rain Water (Q10) last.

Strings

HIGH

Easy-Hard

12 Qs

OBJECTIVE

String algorithm patterns and window-based problems.

SKILLS TESTED

Two pointers, windows, hashing

01	Valid Anagram	E
02	Valid Palindrome	E
03	Longest Substring Without Repeating Characters	M
04	Longest Palindromic Substring	M
05	Group Anagrams	M
06	String to Integer (atoi)	M
07	Longest Common Prefix	E
08	Implement strStr() / substring search	E
09	Minimum Window Substring	H
10	Valid Parentheses	E
11	Roman to Integer	E
12	Count and Say	M

RECOMMENDED ORDER Q9 (Min Window Substring) is hard — attempt after windows.

HashMap

VERY HIGH

Easy-Medium

10 Qs

OBJECTIVE

Hashing-based lookups, counting and grouping.

SKILLS TESTED

HashMap, prefix-sum maps, frequency

01	Two Sum using HashMap	E
02	Subarray Sum Equals K	M
03	Longest Consecutive Sequence	M
04	Group Anagrams	M
05	Top K Frequent Elements	M
06	First Unique Character in a String	E
07	Find All Duplicates in an Array	M
08	Isomorphic Strings	E
09	4Sum Count	M
10	Longest Substring Without Repeating Characters	M

RECOMMENDED ORDER Q2 (Subarray Sum = K) is the key pattern to internalise.

Stack

HIGH

Easy-Medium

10 Qs

OBJECTIVE

Monotonic stacks and expression evaluation.

SKILLS TESTED

Stack, monotonic stack, recursion

01	Valid Parentheses	E
02	Min Stack	M
03	Next Greater Element	M
04	Daily Temperatures	M
05	Evaluate Reverse Polish Notation	M
06	Largest Rectangle in Histogram	H
07	Implement a stack using two queues	M
08	Sort a stack using recursion	M
09	Remove adjacent duplicates from a string	E
10	Trapping Rain Water (stack approach)	H

RECOMMENDED ORDER Q6 & Q10 (histogram, rain water) are the hard finishers.

Queue

MEDIUM

Easy-Medium

8 Qs

OBJECTIVE

Queue, deque and BFS-driven problems.

SKILLS TESTED

Queue, deque, BFS

01	Implement a queue using two stacks	M
02	Implement a circular queue	M
03	Sliding Window Maximum (deque)	H
04	First negative integer in every window of size K	M
05	Generate binary numbers 1..N using a queue	E
06	Rotten Oranges (BFS)	M
07	Number of Islands (BFS)	M
08	Implement an LRU cache	H

RECOMMENDED ORDER Q3 (Sliding Window Max) and Q8 (LRU) are the tough ones.

Linked List

VERY HIGH

Easy-Hard

12 Qs

OBJECTIVE

Pointer manipulation and cycle detection.

SKILLS TESTED

Two pointers, reversal, Floyd's cycle

01	Reverse a linked list (iterative & recursive)	E
02	Detect a cycle in a linked list	M
03	Find the starting node of a cycle	M
04	Find the middle of a linked list	E
05	Merge two sorted linked lists	E
06	Remove the Nth node from the end	M
07	Add two numbers represented by linked lists	M
08	Check if a linked list is a palindrome	M
09	Reverse nodes in K-groups	H
10	Detect and remove a loop	M
11	Find the intersection point of two linked lists	M
12	Flatten a multilevel linked list	H

RECOMMENDED ORDER Q1-Q6 first; Reverse in K-groups (Q9) is the hard one.

Trees

VERY HIGH

Medium-Hard

12 Qs

OBJECTIVE

Traversals, views and structural tree problems.

SKILLS TESTED

DFS, BFS, recursion, BST properties

01	Inorder, Preorder and Postorder traversal	E
02	Level Order traversal (BFS)	M
03	Maximum depth / height of a binary tree	E
04	Check if a tree is height-balanced	M
05	Lowest Common Ancestor	M
06	Diameter of a binary tree	M
07	Validate a Binary Search Tree	M
08	Invert / mirror a binary tree	E
09	Zigzag level order traversal	M
10	Serialize and deserialize a binary tree	H
11	Left and right view of a binary tree	M
12	Convert a sorted array to a balanced BST	M

RECOMMENDED ORDER Traversals first; Serialize/Deserialize (Q10) is the stretch.

Searching

MEDIUM

Easy-Medium

6 Qs

OBJECTIVE

Search variants beyond plain linear scans.

SKILLS TESTED

Binary search variants, matrix search

01	Linear search	E
02	Search in a rotated sorted array	M
03	First and last position of an element	M
04	Search in a 2D matrix	M
05	Find a peak element	M
06	Search in an infinite sorted array	M

RECOMMENDED ORDER Solve in order; Q2 & Q4 are the common interview picks.

Sorting

MEDIUM

Easy-Medium

8 Qs

OBJECTIVE

Implement and apply classic sorting algorithms.

SKILLS TESTED

Merge/quick sort, partitioning, heaps

01 Implement Bubble, Selection and Insertion sort

E

02 Implement Merge Sort

M

03 Implement Quick Sort

M

04 Sort Colors (Dutch National Flag)

M

05 Sort an array of 0s, 1s and 2s

M

06 Merge intervals after sorting

M

07 Find the Kth largest element

M

08 Sort a nearly sorted (K-sorted) array

M

RECOMMENDED ORDER Implement Q1-Q3, then apply to Q4-Q8.

Binary Search

HIGH

Medium

8 Qs

OBJECTIVE

Binary search on arrays and on the answer space.

SKILLS TESTED

Boundaries, search-on-answer

01	Classic binary search	E
02	First and last occurrence of a target	M
03	Search in a rotated sorted array	M
04	Integer square root of a number	M
05	Minimum in a rotated sorted array	M
06	Median of two sorted arrays	H
07	Koko Eating Bananas (search on answer)	M
08	Allocate minimum pages / ship within D days	H

RECOMMENDED ORDER Q6 (Median of two sorted arrays) is hard — leave for last.

Sliding Window

HIGH

Medium

8 Qs

OBJECTIVE

Fixed and variable window techniques.

SKILLS TESTED

Windows, frequency maps

01	Maximum sum subarray of size K	E
02	Longest substring without repeating characters	M
03	Longest substring with K distinct characters	M
04	Minimum window substring	H
05	Maximum of all subarrays of size K	H
06	Count anagrams of a pattern in a string	M
07	Fruit Into Baskets	M
08	Longest repeating character replacement	M

RECOMMENDED ORDER Q4 & Q5 (min window, window max) are the hard finishers.

Two Pointer

HIGH

Easy-Medium

8 Qs

OBJECTIVE

Opposite-end and same-direction pointer patterns.

SKILLS TESTED

Sorted-array pointers, partitioning

01	Pair with a given sum in a sorted array	E
02	3Sum	M
03	Container With Most Water	M
04	Remove duplicates from a sorted array	E
05	Trapping Rain Water	H
06	Sort Colors	M
07	Move Zeroes	E
08	Valid Palindrome (alphanumeric only)	E

RECOMMENDED ORDER Q5 (Trapping Rain Water) is the hard one.

Recursion

MEDIUM-HIGH

Medium-Hard

10 Qs

OBJECTIVE

Recursion and backtracking foundations.

SKILLS TESTED

Recursion, backtracking, pruning

01	Factorial and Fibonacci using recursion	E
02	Tower of Hanoi	M
03	Generate all subsets (power set)	M
04	Generate all permutations of a string/array	H
05	N-Queens	H
06	Rat in a Maze	H
07	Sudoku Solver	H
08	Generate all balanced parentheses	M
09	Subset Sum / Combination Sum	M
10	Word Search in a grid	H

RECOMMENDED ORDER Subsets/permutations first; Sudoku (Q7) and Word Search (Q10) last.

PRIORITY LIST

100

Top 100 Must-Solve Questions

Solve every one of these before your first interview.

1	Two Sum	16	Valid Parentheses
2	Best Time to Buy and Sell Stock	17	Minimum Window Substring
3	Maximum Subarray (Kadane's)	18	Subarray Sum Equals K
4	Product of Array Except Self	19	Longest Consecutive Sequence
5	Find the Duplicate Number	20	Top K Frequent Elements
6	Merge Intervals	21	Min Stack
7	Container With Most Water	22	Next Greater Element
8	Trapping Rain Water	23	Daily Temperatures
9	Set Matrix Zeroes	24	Largest Rectangle in Histogram
10	Spiral Matrix	25	Queue using two stacks
11	Valid Anagram	26	Sliding Window Maximum
12	Valid Palindrome	27	Number of Islands (BFS)
13	Longest Substring Without Repeating Characters	28	Rotten Oranges
14	Longest Palindromic Substring	29	Reverse a linked list
15	Group Anagrams	30	Detect a cycle (start node)

31 Merge two sorted lists

32 Remove Nth node from end

33 Linked list palindrome

34 Reverse nodes in K-groups

35 Intersection of two lists

36 Tree traversals (in/pre/post)

37 Level order traversal

38 Balanced tree check

39 Lowest Common Ancestor

40 Validate BST

41 Serialize / deserialize a tree

42 Search in rotated sorted array

43 Median of two sorted arrays

44 Kth largest element

45 Koko Eating Bananas

46 3Sum

47 Longest repeating char replacement

48 Subsets / permutations

49 N-Queens

50 Word Search

51 Prime check / primes up to N

52 Reverse an integer

53 Number palindrome

54 Fibonacci series

55 Armstrong number

56 GCD and LCM

57 Digital root

58 Power of two check

59 Pascal's triangle

60 Diamond pattern

61 Reverse a string (no built-in)

62 String palindrome

63 First non-repeating character

64 Check anagrams

65 Remove duplicate characters

66 Word frequency in a sentence

67 Reverse words in a sentence

68 String compression (a3b2)

69 String rotation check

70 All permutations of a string

71 Second largest in array

72 Remove duplicates from array

73 Missing number 1..N

74 Rotate array by K

75 Kadane's maximum subarray

76 Move zeros to end

77 Pairs with a given sum

78 Merge two sorted arrays

79 Frequency map with HashMap

80 Sort HashMap by value

81 LRU cache (LinkedHashMap)

82 Top K frequent elements

83 Custom Comparator sort

84 Streams: groupingBy frequency

85 Streams: second highest number

86 Streams: group employees by department

87 Streams: average per department

88 Streams: partition even/odd

89 Streams: highest-paid per department

90 Streams: reduce for sum

91 Immutable class design

92 Thread-safe Singleton

93 Factory pattern

94 equals() and hashCode()

95 Custom exception (checked/unchecked)

96 try-with-resources

97 Odd-even printing with two threads

98 Producer-consumer (wait/notify)

99 Callable and Future

100 Print 1..N with N threads

Top 50 Java Questions

1 Prime check / primes up to N

2 Reverse an integer

3 Number palindrome

4 Fibonacci series

5	Armstrong number	28	Merge two sorted arrays
6	GCD and LCM	29	Frequency map with HashMap
7	Digital root	30	Sort HashMap by value
8	Power of two check	31	LRU cache (LinkedHashMap)
9	Pascal's triangle	32	Top K frequent elements
10	Diamond pattern	33	Custom Comparator sort
11	Reverse a string (no built-in)	34	Streams: groupingBy frequency
12	String palindrome	35	Streams: second highest number
13	First non-repeating character	36	Streams: group employees by department
14	Check anagrams	37	Streams: average per department
15	Remove duplicate characters	38	Streams: partition even/odd
16	Word frequency in a sentence	39	Streams: highest-paid per department
17	Reverse words in a sentence	40	Streams: reduce for sum
18	String compression (a3b2)	41	Immutable class design
19	String rotation check	42	Thread-safe Singleton
20	All permutations of a string	43	Factory pattern
21	Second largest in array	44	equals() and hashCode()
22	Remove duplicates from array	45	Custom exception (checked/unchecked)
23	Missing number 1..N	46	try-with-resources
24	Rotate array by K	47	Odd-even printing with two threads
25	Kadane's maximum subarray	48	Producer-consumer (wait/notify)
26	Move zeros to end	49	Callable and Future
27	Pairs with a given sum	50	Print 1..N with N threads

Top 50 DSA Questions

1	Two Sum	15	Group Anagrams
2	Best Time to Buy and Sell Stock	16	Valid Parentheses
3	Maximum Subarray (Kadane's)	17	Minimum Window Substring
4	Product of Array Except Self	18	Subarray Sum Equals K
5	Find the Duplicate Number	19	Longest Consecutive Sequence
6	Merge Intervals	20	Top K Frequent Elements
7	Container With Most Water	21	Min Stack
8	Trapping Rain Water	22	Next Greater Element
9	Set Matrix Zeroes	23	Daily Temperatures
10	Spiral Matrix	24	Largest Rectangle in Histogram
11	Valid Anagram	25	Queue using two stacks
12	Valid Palindrome	26	Sliding Window Maximum
13	Longest Substring Without Repeating Characters	27	Number of Islands (BFS)
14	Longest Palindromic Substring	28	Rotten Oranges

29 Reverse a linked list

30 Detect a cycle (start node)

31 Merge two sorted lists

32 Remove Nth node from end

33 Linked list palindrome

34 Reverse nodes in K-groups

35 Intersection of two lists

36 Tree traversals (in/pre/post)

37 Level order traversal

38 Balanced tree check

39 Lowest Common Ancestor

40 Validate BST

41 Serialize / deserialize a tree

42 Search in rotated sorted array

43 Median of two sorted arrays

44 Kth largest element

45 Koko Eating Bananas

46 3Sum

47 Longest repeating char replacement

48 Subsets / permutations

49 N-Queens

50 Word Search



TARGET SMART

Company-wise Coding Priority

Where to focus for each employer.

Amazon

Problem-solving depth + Leadership Principles.

PRIORITY TOPICS

Arrays

HashMap

Trees

BFS/Graphs

Sliding Window

DP intro

SIGNATURE QUESTIONS

Two Sum

LRU Cache

Number of Islands

Merge Intervals

Top K Frequent

Microsoft

Pointers, trees and clean coding.

PRIORITY TOPICS

Linked List

Trees

Strings

Arrays

Recursion

SIGNATURE QUESTIONS

Reverse a linked list

Lowest Common Ancestor

Serialize a tree

Clone complex list

Adobe

Arrays, strings and applied logic.

PRIORITY TOPICS

Arrays

Strings

HashMap

Two Pointer

DP intro

SIGNATURE QUESTIONS

Kadane's

Longest substring

Group Anagrams

Trapping Rain Water

Salesforce

Strong Java + OOP + Collections.

PRIORITY TOPICS

OOP Coding

Collections

Strings

HashMap

SIGNATURE QUESTIONS

LRU Cache

Anagrams

OOP design (parking lot)

Sort HashMap by value

Atlassian

Practical coding and applied design.

PRIORITY TOPICS

Collections HashMap Strings Design-lite

SIGNATURE QUESTIONS

Rate limiter design LRU Cache Word frequency
String parsing

Walmart Global Tech

Arrays, trees and streams.

PRIORITY TOPICS

Arrays Trees Sliding Window Java 8 Streams

SIGNATURE QUESTIONS

Sliding Window Maximum Kadane's Tree traversals
Streams grouping

Deloitte

Core Java + fundamentals.

PRIORITY TOPICS

Core Java Collections Java 8 Streams Basic DSA

SIGNATURE QUESTIONS

String/array logic Collections frequency
Streams grouping Number logic

EY

Java logic and OOP basics.

PRIORITY TOPICS

Core Java OOP Coding Collections Logic

SIGNATURE QUESTIONS

Number & string logic OOP coding Collections
Pattern printing

PwC

Java fundamentals and reasoning.

PRIORITY TOPICS

Core Java Collections Java 8 Streams Logic

SIGNATURE QUESTIONS

String/array logic Streams basics Collections
Palindrome/anagram

KPMG

Logical and pattern-based coding.

PRIORITY TOPICS

Core Java Patterns Collections Basic DSA

SIGNATURE QUESTIONS

Pattern printing Number logic Collections
Array basics

Coding Tracker

Log solved and revised counts per topic. Aim for 5★ confidence before moving on.

Section	Topic	Total Qs	Solved	Revised	Confidence
Java	Numbers	15			☆☆☆☆☆
Java	Patterns	10			☆☆☆☆☆
Java	Strings	15			☆☆☆☆☆
Java	Arrays	15			☆☆☆☆☆
Java	Collections	12			☆☆☆☆☆
Java	Java 8 Streams	14			☆☆☆☆☆
Java	OOP Coding	10			☆☆☆☆☆
Java	Exception Handling	8			☆☆☆☆☆

Section	Topic	Total Qs	Solved	Revised	Confidence
Java	Multithreading	10			☆ ☆ ☆ ☆ ☆
DSA	Arrays	15			☆ ☆ ☆ ☆ ☆
DSA	Strings	12			☆ ☆ ☆ ☆ ☆
DSA	HashMap	10			☆ ☆ ☆ ☆ ☆
DSA	Stack	10			☆ ☆ ☆ ☆ ☆
DSA	Queue	8			☆ ☆ ☆ ☆ ☆
DSA	Linked List	12			☆ ☆ ☆ ☆ ☆
DSA	Trees	12			☆ ☆ ☆ ☆ ☆
DSA	Searching	6			☆ ☆ ☆ ☆ ☆
DSA	Sorting	8			☆ ☆ ☆ ☆ ☆
DSA	Binary Search	8			☆ ☆ ☆ ☆ ☆
DSA	Sliding Window	8			☆ ☆ ☆ ☆ ☆
DSA	Two Pointer	8			☆ ☆ ☆ ☆ ☆
DSA	Recursion	10			☆ ☆ ☆ ☆ ☆

Revision Tracker

Re-touch every topic at least twice. Below 4★ means schedule another pass.

Topic	Last Solved	Attempts	Confidence	Next Revision
Java • Numbers			☆ ☆ ☆ ☆ ☆	
Java • Patterns			☆ ☆ ☆ ☆ ☆	
Java • Strings			☆ ☆ ☆ ☆ ☆	
Java • Arrays			☆ ☆ ☆ ☆ ☆	
Java • Collections			☆ ☆ ☆ ☆ ☆	
Java • Java 8 Streams			☆ ☆ ☆ ☆ ☆	
Java • OOP Coding			☆ ☆ ☆ ☆ ☆	
Java • Exception Handling			☆ ☆ ☆ ☆ ☆	
Java • Multithreading			☆ ☆ ☆ ☆ ☆	
DSA • Arrays			☆ ☆ ☆ ☆ ☆	
DSA • Strings			☆ ☆ ☆ ☆ ☆	
DSA • HashMap			☆ ☆ ☆ ☆ ☆	
DSA • Stack			☆ ☆ ☆ ☆ ☆	
DSA • Queue			☆ ☆ ☆ ☆ ☆	

Topic	Last Solved	Attempts	Confidence	Next Revision
DSA • Linked List			☆ ☆ ☆ ☆ ☆	
DSA • Trees			☆ ☆ ☆ ☆ ☆	
DSA • Searching			☆ ☆ ☆ ☆ ☆	
DSA • Sorting			☆ ☆ ☆ ☆ ☆	
DSA • Binary Search			☆ ☆ ☆ ☆ ☆	
DSA • Sliding Window			☆ ☆ ☆ ☆ ☆	
DSA • Two Pointer			☆ ☆ ☆ ☆ ☆	
DSA • Recursion			☆ ☆ ☆ ☆ ☆	

30-Day Coding Revision Plan

Blue rows are review / mock checkpoints. Daily target is a guide — adjust to your pace.

Day	Focus	Target Qs	Done
Day 1	Java • Numbers + Patterns	10	
Day 2	Java • Strings	12	
Day 3	Java • Arrays	12	
Day 4	Java • Collections	10	
Day 5	Java • Java 8 Streams	12	
Day 6	Java • OOP + Exceptions	12	
Day 7	REVIEW • Java Week 1 + weak spots	10	
Day 8	Java • Multithreading	10	
Day 9	DSA • Arrays	12	
Day 10	DSA • Strings	10	
Day 11	DSA • HashMap	10	
Day 12	DSA • Stack	10	
Day 13	DSA • Queue	8	
Day 14	REVIEW • DSA so far	12	
Day 15	DSA • Linked List	12	
Day 16	DSA • Trees (traversals)	8	
Day 17	DSA • Trees (advanced)	8	
Day 18	DSA • Searching + Sorting	12	
Day 19	DSA • Binary Search	8	
Day 20	DSA • Sliding Window	8	

Day	Focus	Target Qs	Done
Day 21	REVIEW • Weeks 2-3 weak spots	12	
Day 22	DSA • Two Pointer	8	
Day 23	DSA • Recursion	10	
Day 24	MIXED • timed set (Easy→Medium)	10	
Day 25	MIXED • timed set (Medium→Hard)	8	
Day 26	Top 50 Java • re-solve	15	
Day 27	Top 50 DSA • re-solve	15	
Day 28	Company-focused set (target company)	12	
Day 29	Weak-topic blitz	12	
Day 30	FULL MOCK • 10 mixed problems, timed	10	

Don't just read them. Solve them — twice.

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